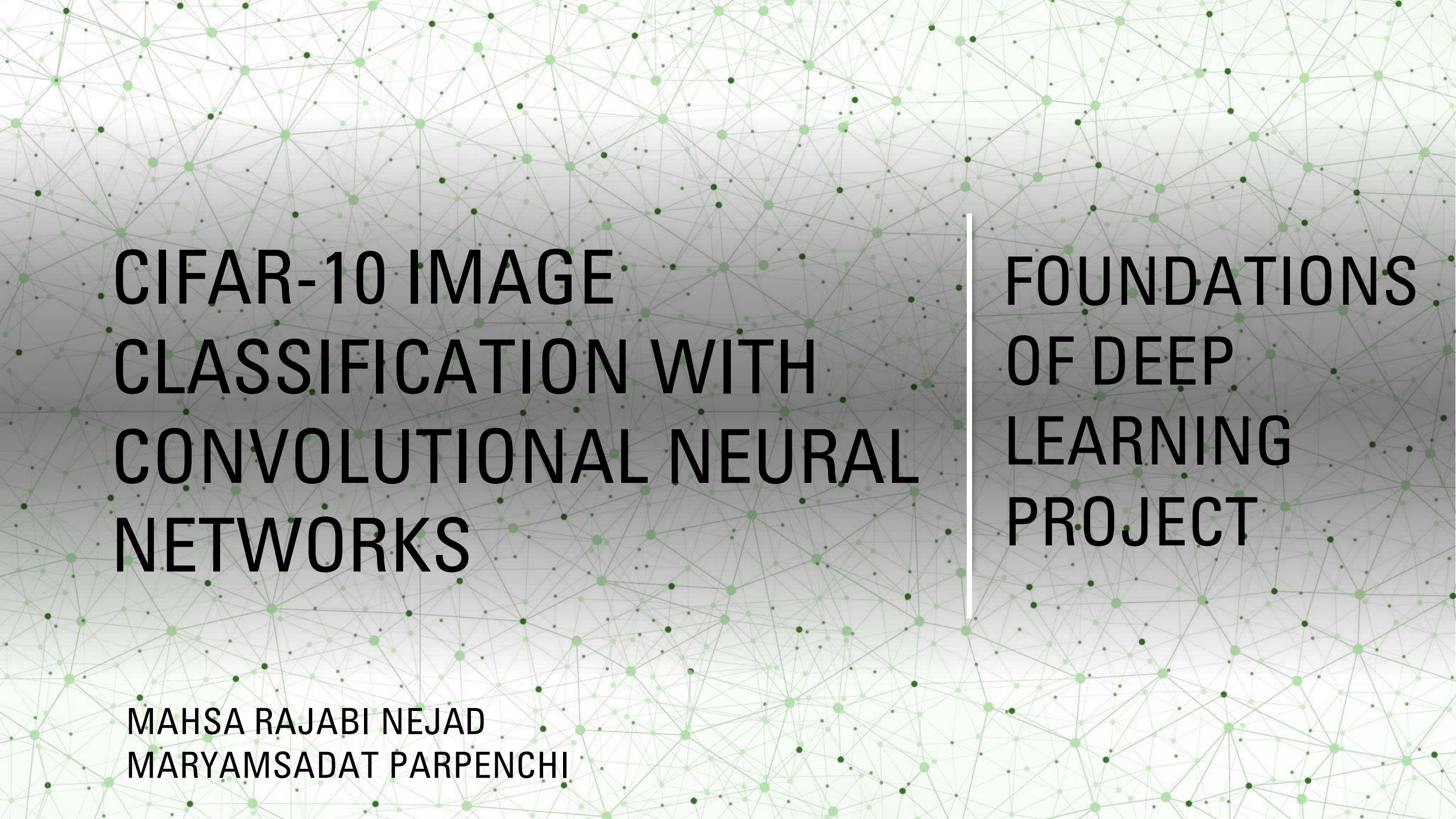


The background of the slide is a complex network of green and black dots connected by thin lines, creating a web-like pattern. The dots are of varying sizes and colors, with some being bright green and others dark green or black. The lines are thin and greyish-green.

CIFAR-10 IMAGE CLASSIFICATION WITH CONVOLUTIONAL NEURAL NETWORKS

FOUNDATIONS OF DEEP LEARNING PROJECT

MAHSA RAJABI NEJAD
MARYAMSADAT PARPENCHI

PROBLEM DEFINITION

- Task: Multi-class image classification
 - Input: 32×32 RGB image
 - Output: One of 10 classes
 - Model type: Convolutional Neural Network
-
- Dataset source: Official CIFAR-10 website, University of Toronto
<https://www.cs.toronto.edu/~kriz/cifar.html>
-

DATASET DESCRIPTION

DATASET: CIFAR-10

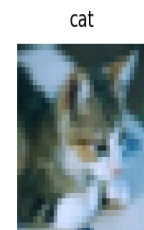
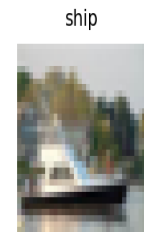
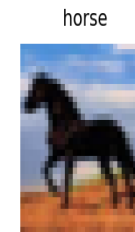
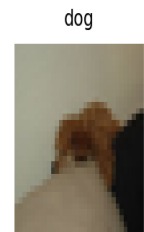
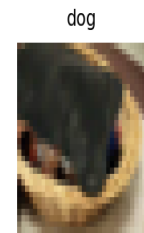
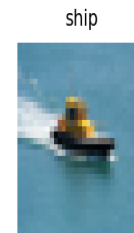
TOTAL IMAGES: 60,000

TRAINING IMAGES: 50,000

TEST IMAGES: 10,000

CLASSES: 10 BALANCED
CLASSES

IMAGE SIZE: $32 \times 32 \times 3$



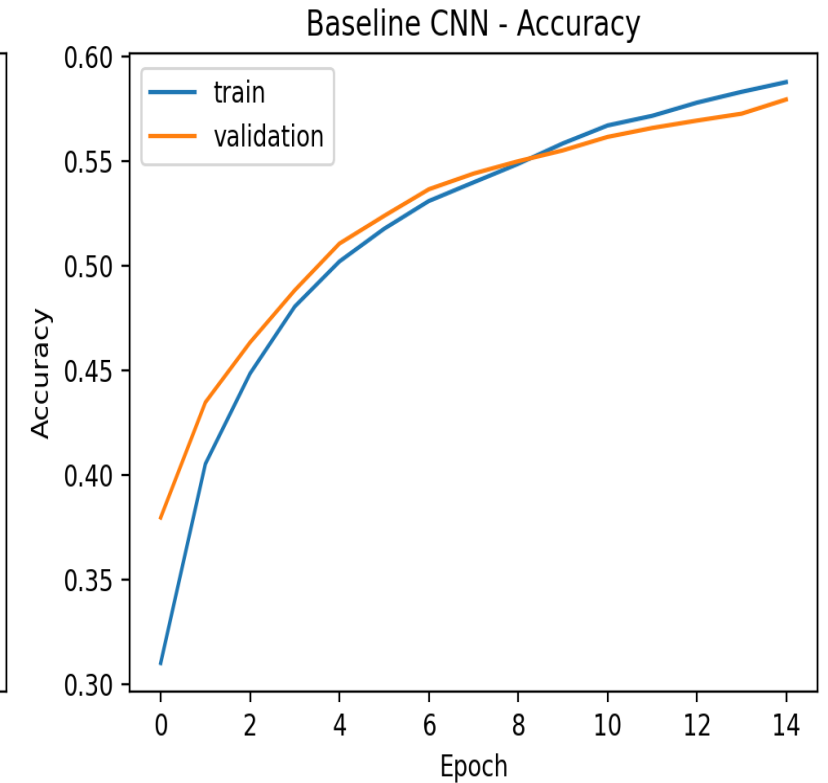
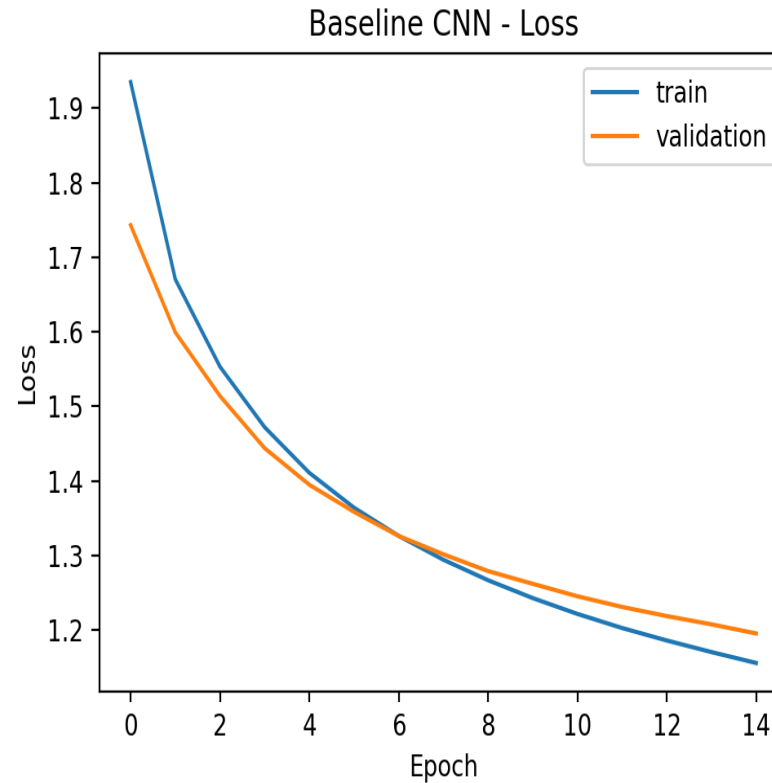
DATA PREPROCESSING

- Loaded MATLAB .mat batches
 - Reshaped images: $3072 \rightarrow 32 \times 32 \times 3$
 - Normalized pixels: $[0, 255] \rightarrow [0, 1]$
 - Training split:
 - 40,000 train
 - 10,000 validation
 - Official test set:
 - 10,000 test images
 - Labels converted to one-hot encoding
-

BASELINE CNN

- Architecture:
- Conv2D(32)
- ReLU
- Conv2D(64)
- ReLU
- GlobalMaxPooling2D
- Dense(10, softmax)

- Test accuracy: 57.49%
- Test loss: 1.1902
- Parameters: 20,042



IMPROVED CNN

Main improvements:

More convolutional layers

MaxPooling2D

BatchNormalization

Dropout

Dense hidden layer

Adam optimizer

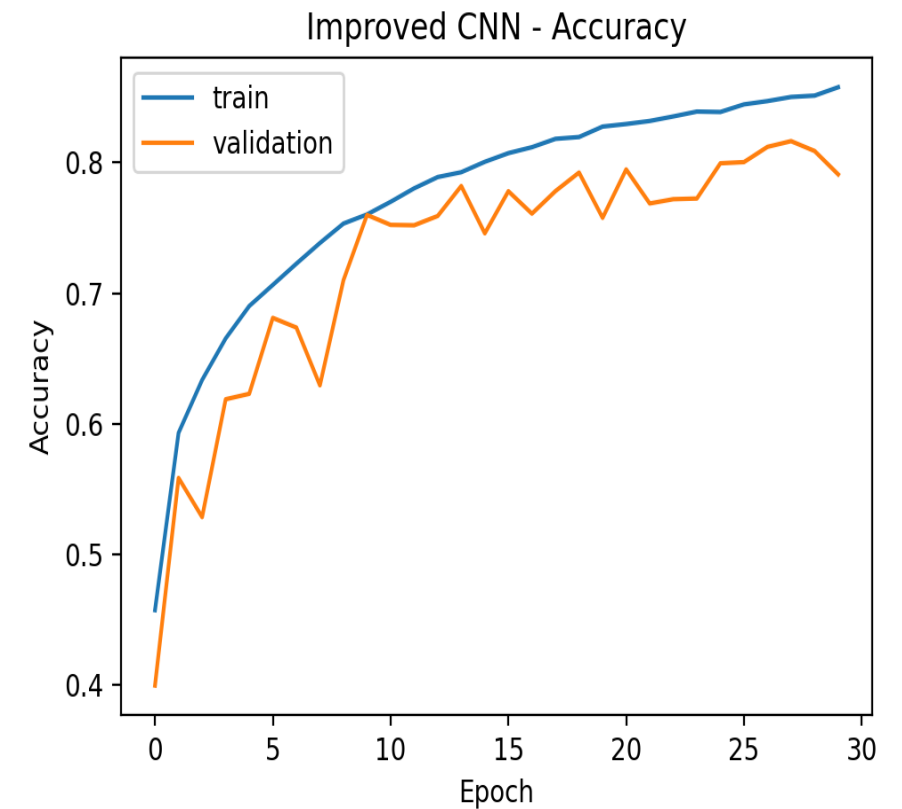
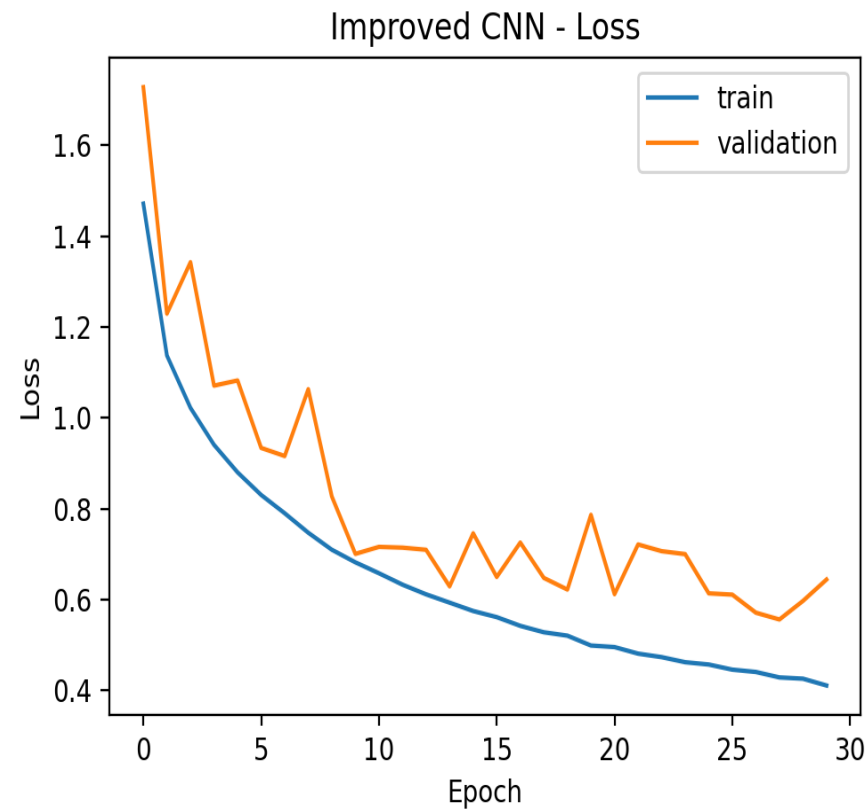
EarlyStopping

ModelCheckpoint

Test accuracy: 81.09%

Test loss: 0.5705

Parameters: 158,506



DATA AUGMENTATION EXPERIMENT

Augmentation methods:

RandomFlip

RandomRotation

RandomZoom

RandomTranslation

RandomContrast

Test accuracy: 76.69%

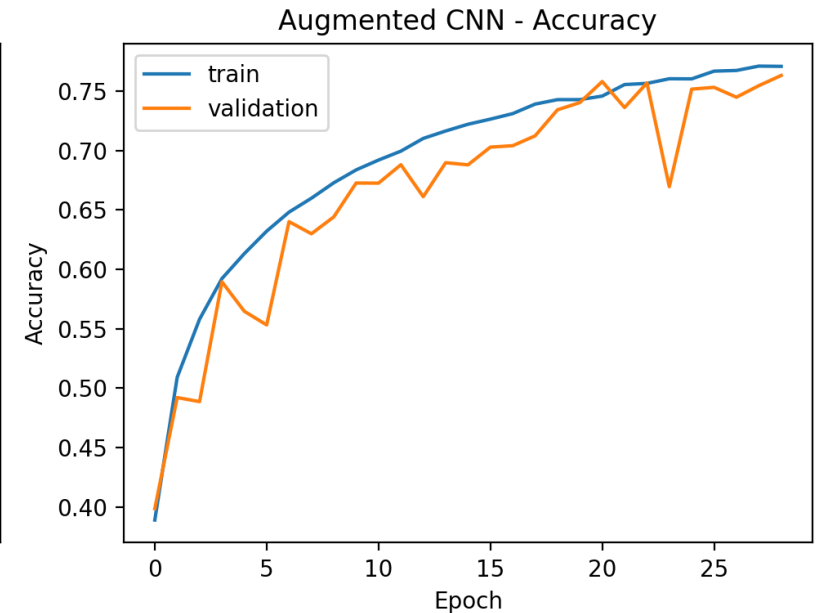
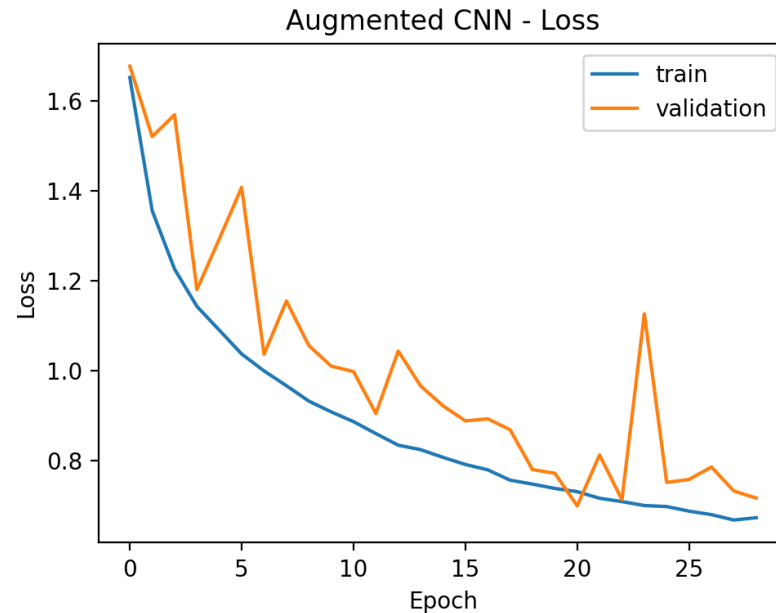
Test loss: 0.6880

Observation:

Accuracy decreased compared to Improved CNN.

Possible explanation:

Transformations may be too strong for very small 32×32 images.



VALIDATION FLUCTUATION AND LR SCHEDULER

Problem:

Improved CNN showed validation fluctuations.

Possible reasons:

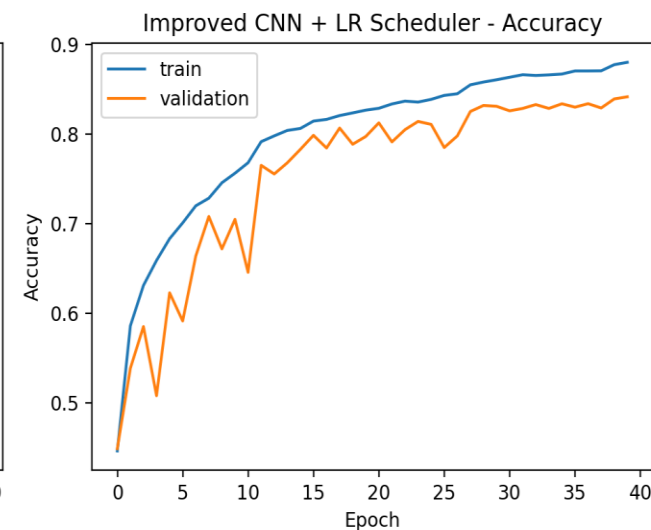
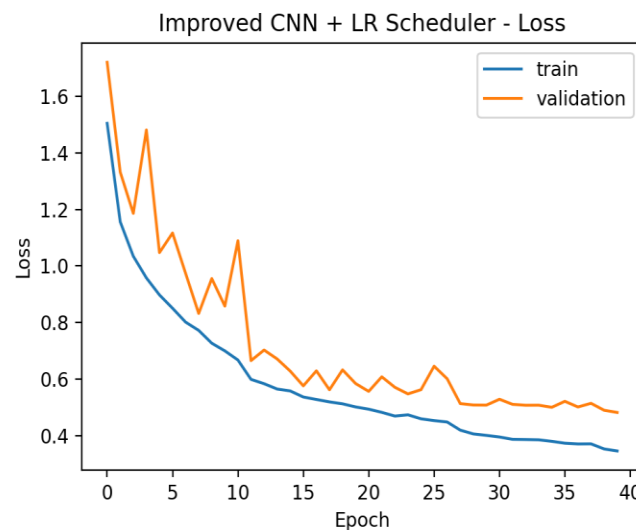
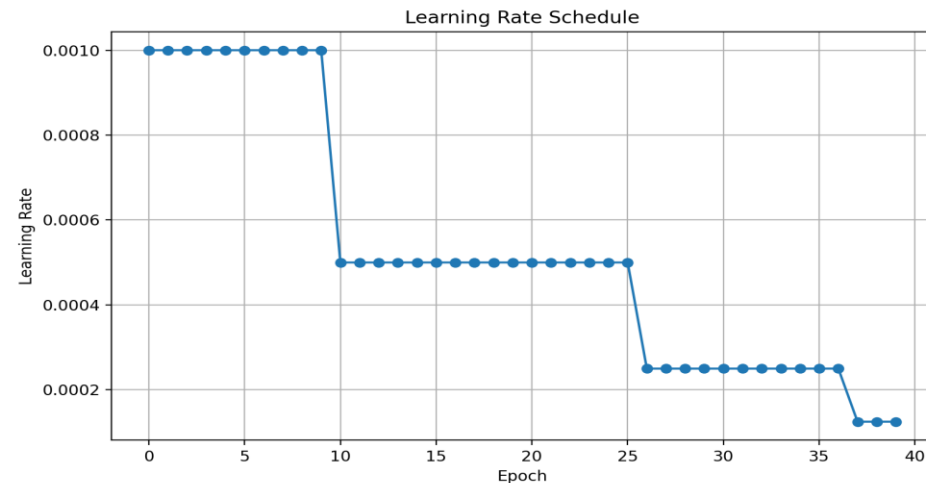
- Mini-batch training introduces noisy updates
- Dropout and Batch Normalization add stochastic behavior
- Fixed learning rate may be too large in later epochs

Solution:

ReduceLROnPlateau

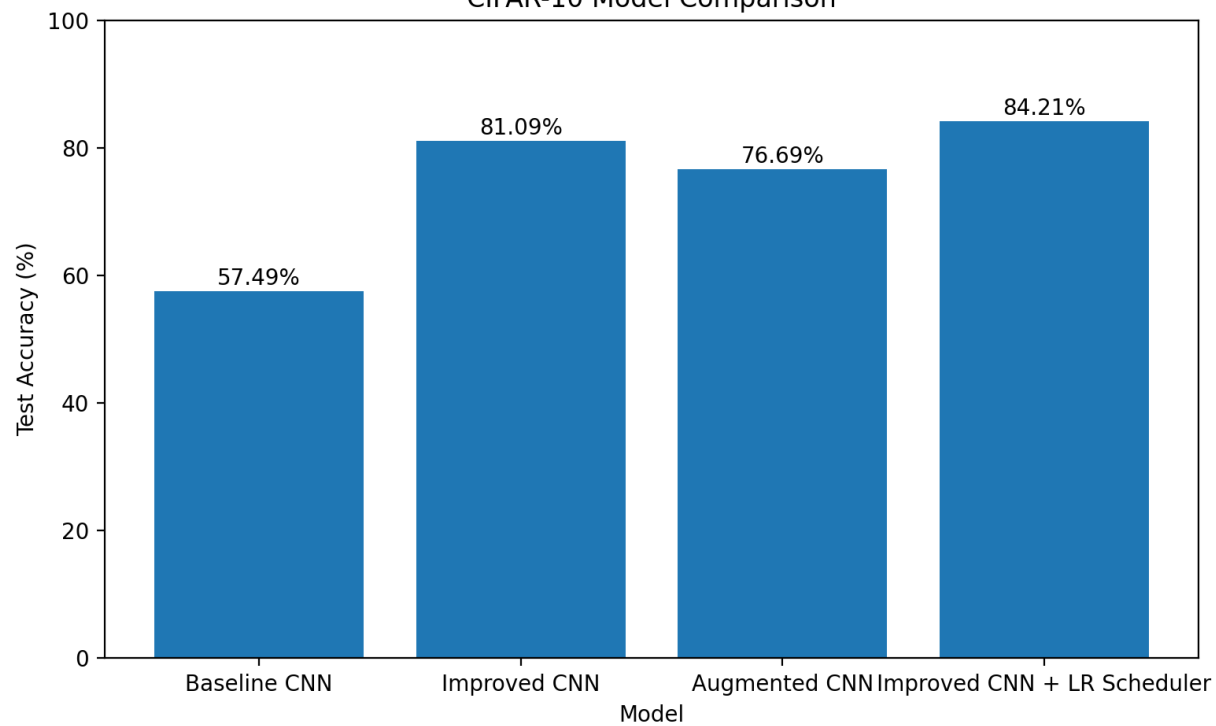
Learning rate schedule:

$0.001 \rightarrow 0.0005 \rightarrow 0.00025 \rightarrow 0.000125$

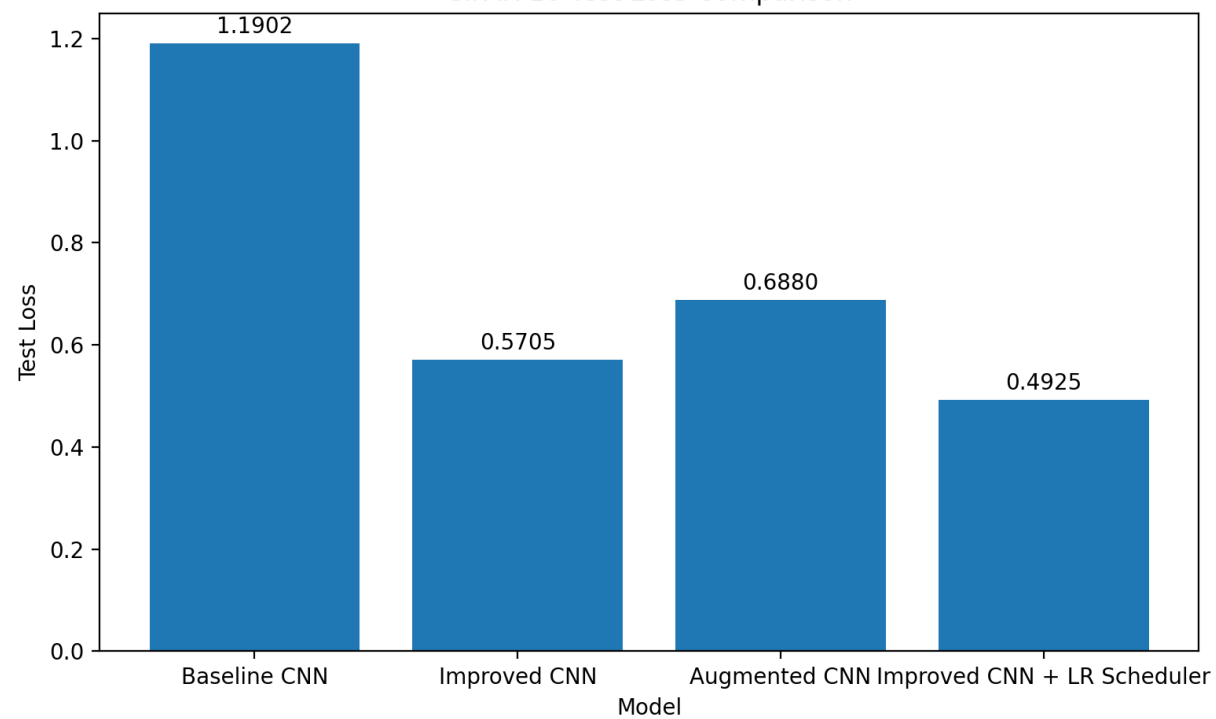


MODEL COMPARISON

CIFAR-10 Model Comparison



CIFAR-10 Test Loss Comparison



MODEL COMPARISON

Model	Test Accuracy	Test Loss	Interpretation
Baseline CNN	57.49%	1.1902	Underfitting
Improved CNN	81.09%	0.5705	Stronger architecture
Augmented CNN	76.69%	0.6880	Augmentation too strong
Improved CNN + LR Scheduler	84.21%	0.4925	Best model

FINAL SELECTED MODEL

Model : Improved CNN + ReduceLROnPlateau

Test accuracy: 84.21%

Test loss: 0.4925

Best validation accuracy: 84.17%

Macro F1-score: 84.14%

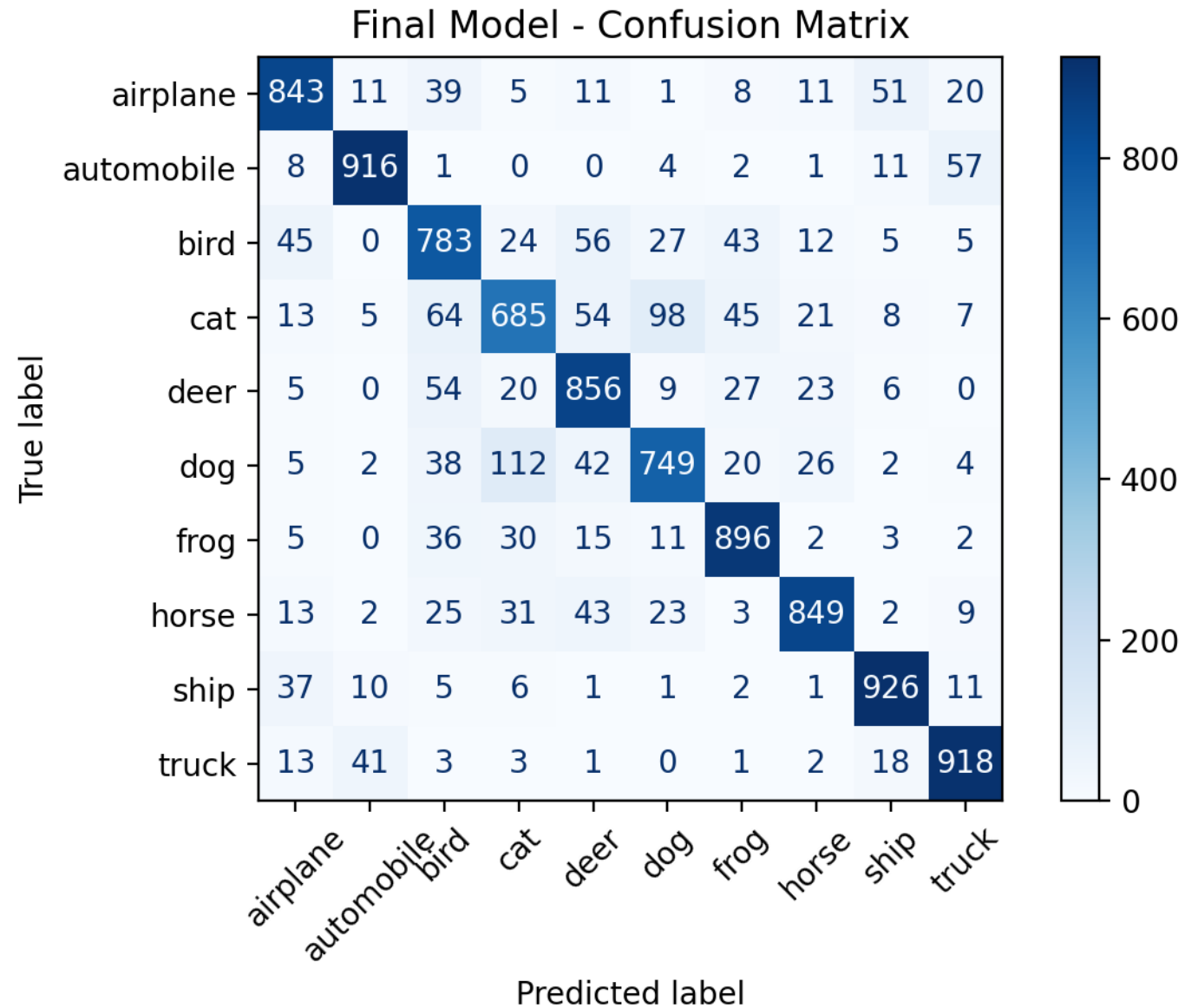
Parameters: 158,506

Saved model: improved_cnn_lr_scheduler_best.keras

CONFUSION MATRIX ANALYSIS

Main observations:

- Best classes: automobile, ship, truck, frog
- Harder classes: cat, bird, dog
- Animal classes are more frequently confused with each other
- Vehicle-like classes are easier to distinguish



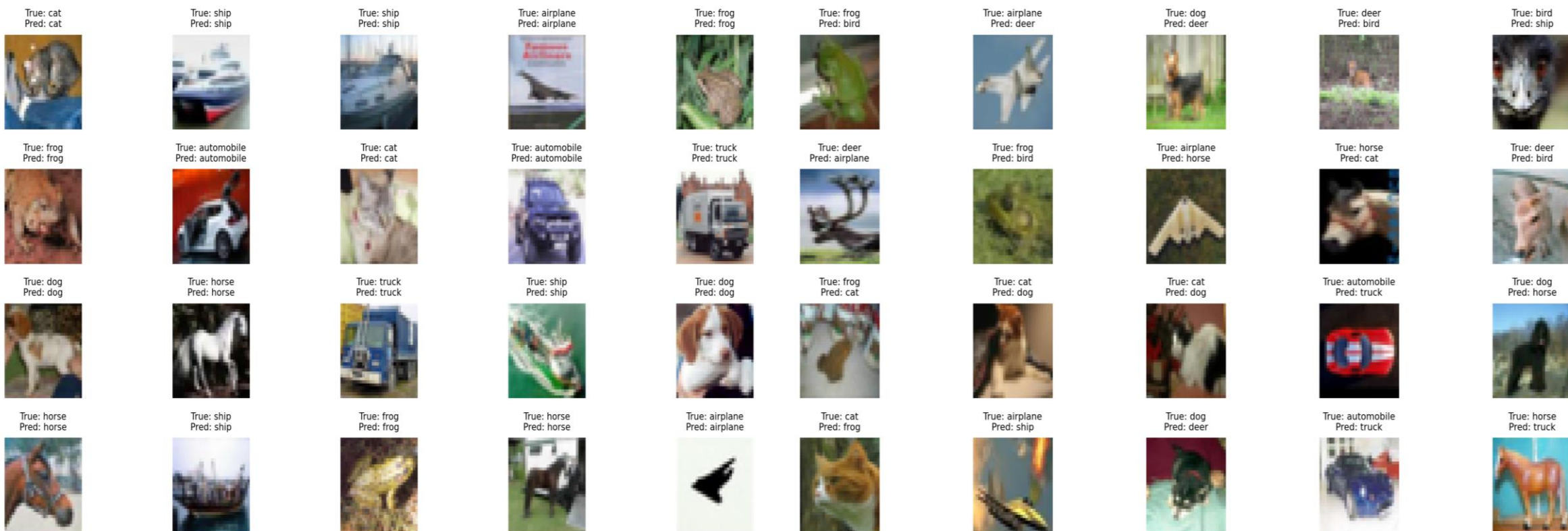
CORRECT AND WRONG PREDICTIONS

Correct predictions:

- Clear object shape
- Distinct visual pattern
- Easier classes such as ship, truck, automobile, frog

Wrong predictions:

- Low-resolution ambiguity
- Similar animal classes
- Unclear object boundaries



CONCLUSIONS

- CNNs are effective for CIFAR-10 classification.
 - Increasing model capacity significantly improved performance.
 - BatchNormalization and Dropout helped regularization.
 - Strong data augmentation was not beneficial for 32×32 images.
 - ReduceLROnPlateau improved convergence and final accuracy.
 - Final test accuracy: 84.21%
 - The final model achieved the best performance by improving the training strategy, not only by increasing architecture complexity.
-

FUTURE WORK

- Try milder data augmentation
 - Add L2 regularization
 - Tune dropout values
 - Use learning-rate scheduling with more epochs
 - Test deeper CNN architectures
 - Evaluate transfer learning if suitable
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